

XGraphCDS: An explainable deep learning model for predicting drug sensitivity from gene pathways and chemical structures

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ARTICLE INFO

Keywords:

Cancer drug sensitivity prediction
Deep learning
Gene pathways
Chemical structures
Interpretation

ABSTRACT

Cancer is a highly complex disease characterized by genetic and phenotypic heterogeneity among individuals. In the era of precision medicine, understanding the genetic basis of these individual differences is crucial for developing new drugs and achieving personalized treatment. Despite the increasing abundance of cancer genomics data, predicting the relationship between cancer samples and drug sensitivity remains challenging. In this study, we developed an explainable graph neural network framework for predicting cancer drug sensitivity (XGraphCDS) based on comparative learning by integrating cancer gene expression information and drug chemical structure knowledge. Specifically, XGraphCDS consists of a unified heterogeneous network and multiple sub-networks, with molecular graphs representing drugs and gene enrichment scores representing cell lines. Experimental results showed that XGraphCDS consistently outperformed most state-of-the-art baselines ($R^2 = 0.863$, $AUC = 0.858$). We also constructed a separate *in vivo* prediction model by using transfer learning strategies with *in vitro* experimental data and achieved good predictive power ($AUC = 0.808$). Simultaneously, our framework is interpretable, providing insights into resistance mechanisms alongside accurate predictions. The excellent performance of XGraphCDS highlights its immense potential in aiding the development of selective anti-tumor drugs and personalized dosing strategies in the field of precision medicine.

1. Introduction

Cancer is a highly complex disease, different patients with the same type of cancer may have varying sensitivities to a single drug, leading to significant differences in treatment efficacy [1]. Therefore, when designing anti-cancer drugs, the heterogeneity of cancer and the individual differences in treatment response among patients are usually taken into consideration. In addition, off-target cell toxicity and drug resistance are also important reasons for their failure in clinical trials with 97% of anti-cancer candidate drugs [2,3]. How to more effectively screen and discover selective anti-cancer drugs, predict the sensitivity of individual tumors to drugs, and scientifically guide personalized clinical medication has become an urgent issue in the current development of anti-cancer drugs.

Patient-derived cancer cell lines have long been used as experimental models of tumor biology in cancer research, and drugs that inhibit the growth of cancer cell lines are thought to inhibit the development of the original tumor [4]. In recent years, several large-scale anticancer drug

screening projects have generated a vast number of drug sensitivity profiles for thousands of cancer cell lines, such as Genomics of Drug Sensitivity in Cancer (GDSC) [5], Cancer Cell Line Encyclopedia (CCLE) [6], and The Cancer Genome Atlas (TCGA) [7]. These publicly available datasets not only provide rich information on anticancer drug responses but also offer genetic maps of cancer cell lines underlying individual differences, making it feasible to accurately predict cancer drug sensitivity (CDS).

Indeed, there has been a significant increase in the application of deep learning algorithms in the field of clinical medicine. Deep learning techniques have shown promise in various clinical tasks such as medical image analysis [8,9], disease diagnosis [10], treatment prediction, and patient monitoring. Some computational CDS models have been proposed, which can be classified into two categories. The first is the similarity network-based prediction method, which assumes that structurally similar compounds have similar sensitivity to the same cell line. They treat drug response prediction as a link prediction problem, such as GraphCDR [11], NIHGCN [12], and TSGCNN [13]. The second is

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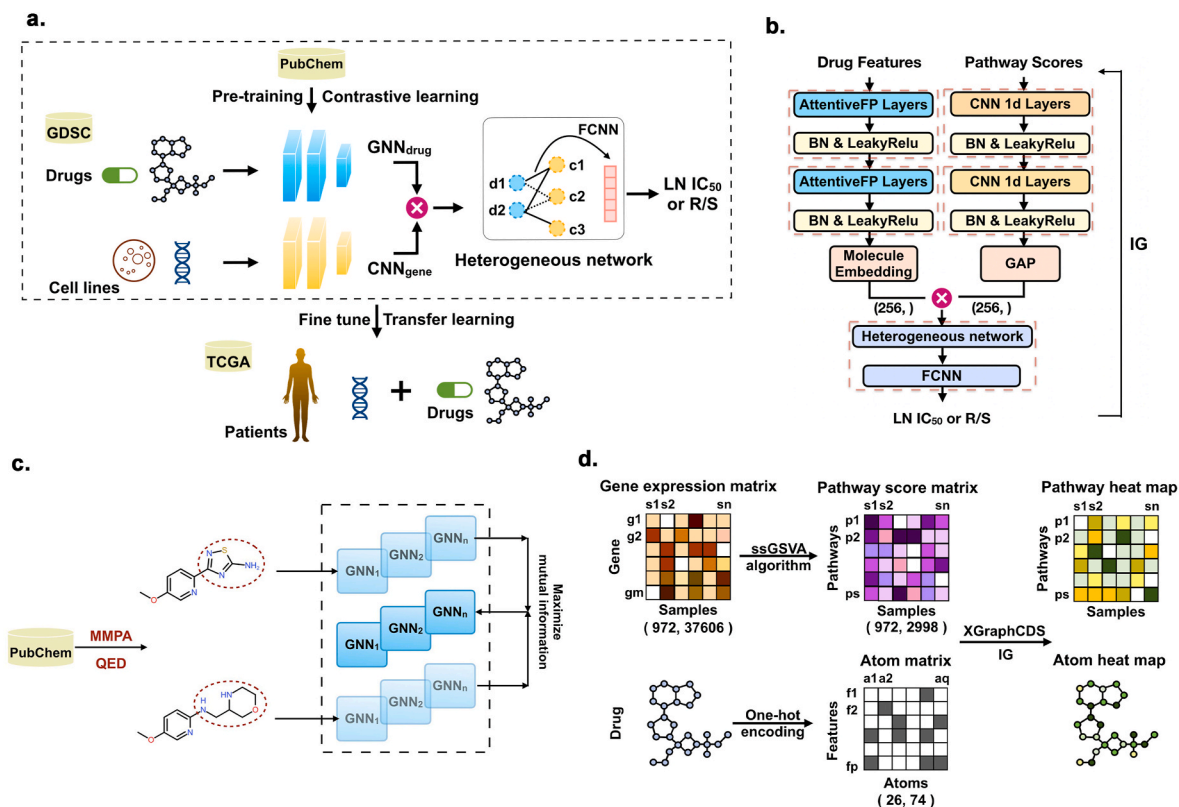


Fig. 1. Description of the predictive workflow. **(a)** Overview of the XGraphCDS network architecture, including three modules GNN_{drug} , CNN_{gene} , and heterogeneous network. **(b)** The details of each XGraphCDS module. The integrated gradients (IG) are obtained by backpropagation for model interpretation. **(c)** Schematic diagram of the pre-training strategy based on contrast learning. **(d)** A pipeline of cell lines and drug representations and interpretations.

the machine learning-based method, which expresses the prediction of cell line-drug response as a regression or classification task. These approaches include traditional machine learning models such as Elastic Network, Random Forest (RF) [14], and Support Vector Machine (SVM) [15], and deep learning models such as DeepCDR [16], PaccMann [17] and Precily [18]. Although these attempts have made significant progress in CDS, they still encounter some complications. Firstly, the recommendation system-based methods often show poor scalability and low computational efficiency [16], and have weak extrapolation ability for new drug-cell line pairs. Secondly, feature selection is a major challenge due to the high-dimensional nature of genomic data. Thirdly, the narrow chemical space of drugs in the dataset restricts the application domain of the model to perform large-scale drug screening. Fourthly, it is difficult to transfer drug response prediction models learned in cell lines (*in vitro*) to the clinical setting *in vivo* [19]. Finally, most models lack interpretability.

To address the limitations of previous work, we integrated genomics and chemoinformatics and proposed an interpretable graph neural network framework (XGraphCDS) for predicting CDS *in vitro* and *in vivo*. This accurate and convenient drug sensitivity prediction model could promote the application of precision cancer medicine in the drug development process, and help researchers design more targeted anti-cancer drugs. In the clinical setting, XGraphCDS is expected to provide clinical decision support for clinicians, aiding in the development of personalized treatment plans.

2. Materials and methods

2.1. Data collection and preparation

2.1.1. Drug responses datasets

The *in vitro* datasets are mainly from the GDSC which provides half-

maximal inhibitory concentration ($LN(IC_{50})$) values for large-scale drug screening data. The dataset mainly includes two projects, GDSC1 and GDSC2. GDSC1 (covering 2010–2015) contains drug screening data from the Wellcome Sanger Institute and Massachusetts General Hospital; GDSC2 (covering 2015 to present) includes the latest screening results from the Wellcome Sanger Institute using improved experimental procedures. We integrated these two datasets, took the intersection for quantitative modeling of drug responses (with GDSC2 data taking precedence for the same data pairs), and took the difference for constructing two external validation sets (old drugs-cell lines and new drugs-cell lines). The old drug-cell line dataset includes some missing pre-existing cell lines and drug-data pairs. The new drugs-cell lines dataset is composed of cell lines or drugs outside of the training set. In addition, we compared the threshold values of each drug provided by the maximum screening concentration reported with the $LN(IC_{50})$ value, and built a binary classification model for drug response, dividing them into sensitivity and resistance [20,21].

The *in vivo* dataset is derived from the TCGA database and contains clinical drug response information for cancer patients. Similar to the *in vitro* data, drug responses were classified into sensitivity (“complete response” and “partial response”) and resistance (“stable disease” and “progressive disease”) based on Chawla et al.’s work [18] using tumor mRNA sequencing data. We also collected the PDX encyclopedia dataset (PDX) in the literature, which includes gene expression profiles and drug response values [22]. PDX drug responses were divided into two groups, sensitivity (“CR” and “PR”) and resistance (“SD” and “PD”). The PDX and TCGA datasets also served as external validations for the GDSC *in vitro* classification model.

To mitigate the risk of information leakage during model construction and training [23], we randomly split the modeling data into a training set and a test set with a ratio of 9:1 using Python scripts. We simultaneously evaluated the model using 5-fold cross-validation (5-CV)

and test set validation. The number of drugs, cell lines, and interaction pairs used in each dataset in this study is listed in Table S1.

2.1.2. Drug molecular representations

We downloaded the SMILES strings of all drugs from the PubChem database [24] based on the drug name and PubChem ID in the GDSC, PDX, and TCGA databases, and preprocessed the chemical structures using RDKit (version 2020.03.1) [25] and MolVS [26], which included standardization, desalting, neutralization, and removal of duplications. In the GNNs, we treated atoms in the compounds as nodes and chemical bonds as edges. We used RDKit and DGL-LifeSci [27] to translate all molecules into molecular graphs by one-hot encoding according to the translation rules (Table S2). We also calculated the ECFP4 fingerprints (1024 bits) [28] of these drugs for model ablation experiments. These implementation details can be found in our related code repository.

2.1.3. Cancer cell line representations

CCLC database [6] provides gene expression data (in TPM values from Array Express) across one thousand cancer cell lines, which can be downloaded directly from the DeMap portal website (<https://depmap.org/>). In this study, we trained our model using RNA-seq gene expression profiles of CCLC cell lines that overlapped with the GDSC dataset cell lines. The raw expression matrix contains 37,606 Ensembl gene IDs, which were converted to official gene symbols and Entrez IDs using the NCBI Gene database [29]. The expression matrix was required to be normalized using $\text{Log}_2(\text{TPM}+1)$. The high-dimensional nature of the genomic feature matrix may not only lead to over-fitting of the model but also increase training costs. Since the majority of targeted therapies act through pathways, we applied the single sample Gene Set Enrichment Analysis (ssGSEA) algorithm [30], developed by the Broad Institute, to perform pathway enrichment analysis on genes. We calculated the GSVA scores based on the c2 collection of canonical pathways of 3050 gene sets selected from the Molecular Signatures Database (MSigDB) [31], setting the min. sz to 5 (removing pathways with fewer than 5 enriched genes) [18]. For the PDX and TCGA datasets, we followed the same procedures to convert gene expression profiles into GSVA pathway scores.

2.2. XGraphCDS framework

2.2.1. Model training protocols

Our deep learning model adopts a dual-convergent architecture, treating drug response prediction as a task of revealing the interactions between drugs and cell lines. As described in Fig. 1a and b, XGraphCDS requires three steps to complete the prediction task. Firstly, the drug molecules are converted into molecular graphs and input into the GNN_{drug} module to extract drug features. At the same time, the enriched gene pathway score matrix is input into the CNN_{gene} module to learn the cell line feature representation, ultimately obtaining drug and cell line attributes with the same dimensions. Then, XGraphCDS inputs the feature vectors of drugs and cell lines into a heterogeneous network and uses a Fully Connected Neural Network (FCNN) to predict their inter-relationship, $\text{LN}(\text{IC}_{50})$. In the GNN_{drug} module, we configured 2 hidden layers with 256 neurons each. Similarly, in the CNN_{gene} module, we also set up 2 layers with a kernel size of 5 and 10 hidden channels.

To avoid overfitting and save computational resources, we set Batch Normalization [32] and Dropout [33] after each convolutional layer. The LeakyReLU function [34] is introduced as a nonlinear activation function after linear transformation. We use MSELoss as the loss function for regression tasks. Due to the heavy positive and negative sample imbalance in our classification dataset, we use Focal Loss [35] as the loss function for the classification task. It is derived from the standard cross-entropy loss, which enables the training model to focus more on the hard-to-classify samples by reducing the weights of the easy-to-classify samples. γ and α are two hyper-parameters of Focal Loss, α is used to address class imbalance, while γ is used to control the weight

of difficult samples. According to the experimental results (Fig. S1), the model achieved the best performance with $\alpha = 3$ and $\gamma = 0.75$. All models are optimized with the Adam optimizer [36] and trained using the Deep Graph Library (DGL) Python package (version 0.9.1) [37] -cuda 11.3 and the DGL-LifeSci extension (version 0.3.0) [27], running on the GPU version of the PyTorch framework (1.12.0) [38]. We used cosine annealing to adjust the learning rate (Fig. S2), and the additional hyperparameter settings and the iteration process of the model are summarized in Table S3 and Fig. S3.

2.2.2. Pre-training strategies

Two pre-training strategies were employed here. On the one hand, we designed a self-supervised pre-training method based on contrastive learning, which optimizes the GNN_{drug} subnetwork by maximizing the mutual information between pairs of graph augmentations. As shown in Fig. 1c, we utilized mmpdb (the matched molecular pair analysis platform) [39] to perform substructure replacements of nearly 10 million small molecules from the PubChem database and fed them in pairs into the GNN_{drug} network. Using NT-Xent [40,41] as the objective function to train the model:

$$\text{Loss}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

where z_i , z_j , and z_k are the feature vectors extracted by GNN_{drug} from positive pairs and negative samples, N is the batch size, τ is the temperature parameter, and $\text{sim}(z_i, z_j)$ represents cosine similarity. During pre-training, GNN_{drug} accumulates chemical knowledge of molecules, and the learned encoders and representations will be fine-tuned for downstream tasks.

On the other hand, we adopted a full parameter transfer approach (Fig. 1a). First, we established a cell-drug sensitivity prediction model based on *in vitro* data (source domain) comprising 187,792 data pairs. Then, based on the parameters of the *in vitro* model, we transferred this model to validate and optimize using *in vivo* data from cancer patients (target domain) comprising 3039 data pairs, creating an independent *in vivo* CDS model.

2.2.3. Drug encoder

Given a drug d (Fig. 1d), its molecular graph representation is denoted as $G_d = (N_d, E_d)$. $N_d \in \mathbb{R}^{A_d \times F_d}$ is its node matrix, $E_d \in \mathbb{R}^{B_d \times D_d}$ represents the adjacency matrix of edges, where A_d denotes the number of atoms, B_d denotes the number of chemical bonds, F_d is the encoded atomic features ($F_d = 78$), and D_d is the encoded chemical bond features ($D_d = 13$). AttentiveFP is a GNN framework that is more suitable for processing compounds, introducing attention mechanisms at both the atomic and molecular levels while performing information propagation through gated recursive units. The Attentive FP framework was originally proposed by Xiong and his colleagues, and tailored for molecular feature extraction [42,43]. In the GNN_{drug} module, we use two AttentiveFP layers for updating atom and bond features, and one readout layer for molecular embedding, which ultimately outputs the drug's feature vector $G'_d \in \mathbb{R}^{1 \times 256}$ to the next stage.

2.2.4. Cancer cell line encoder

Given a cancer cell line c with a series of pathway scores, we encode it as $X_c \in \mathbb{R}^{1 \times P_c}$ ($P_c = 2998$), where P_c represents the pathway GSVA score matrix (Fig. 1d). Then, we learn the latent vectors of gene features in the potential space through two 1D convolutional networks. Due to the redundancy of fully connected layers (FCs) parameters (accounting for about 80% of the entire network parameters) [44], we add a global average pooling layer to replace FCs and integrate the learned deep features, transforming them into $X'_c \in \mathbb{R}^{1 \times 256}$ as input to the heterogeneous network.

2.2.5. Heterogeneous network for drug response prediction

The cancer cell line-drug response can be represented as an undirected graph $G_r = (V_{D+C}, E)$, where V_{D+C} represents the set of nodes of drugs and cancer cell lines, and E represents all edges between them. In the previous steps, we obtained the hidden representations G'_d and X'_c of drugs and cell lines through the GNN_{drug} and CNN_{gene} subnetworks. Now, we construct a heterogeneous network by Hadamard product ($G'_d \odot X'_c$). We use FCNN to predict a score for each pair of nodes (V_D, V_C) that do not have an edge in the network. This score is the drug response value ($LN(IC_{50})$) in the regression task and the probability of the drug being sensitive to the cell line sample in the binary classification task. The pseudo-codes of XGraphCDS are illustrated in **Algorithm 1**.

Algorithm 1: XGraphCDS

Input:

Drug molecular graph $G_d \leftarrow (N_d, E_d)$ Cancer cell line pathway scores matrix $X_c \in \mathbb{R}^{1 \times P_c}$

Output:

Predicted drug response value or probability

 GNN_{drug} : //Drug encoder using AttentiveFPfor i in range(2): $N_d \leftarrow \text{AttentiveFP_layer}(N_d, E_d)$ $E_d \leftarrow \text{AttentiveFP_layer}(E_d, N_d)$ $G'_d \leftarrow \text{readout_layer}(N_d)$ CNN_{gene} : //Encode pathway scores into gene features $X_c \leftarrow \text{GSVA}(X_c)$ for i in range(2): $X_c \leftarrow \text{1D_conv_layer}(X_c)$ $X'_c \leftarrow \text{global_average_pooling}(X_c)$

Heterogeneous_network: //Heterogeneous network for drug response prediction

 $G_r \leftarrow G'_d \odot X'_c$ predicted_value $\leftarrow \text{FCNN_prediction}(G_r)$

return predicted_value

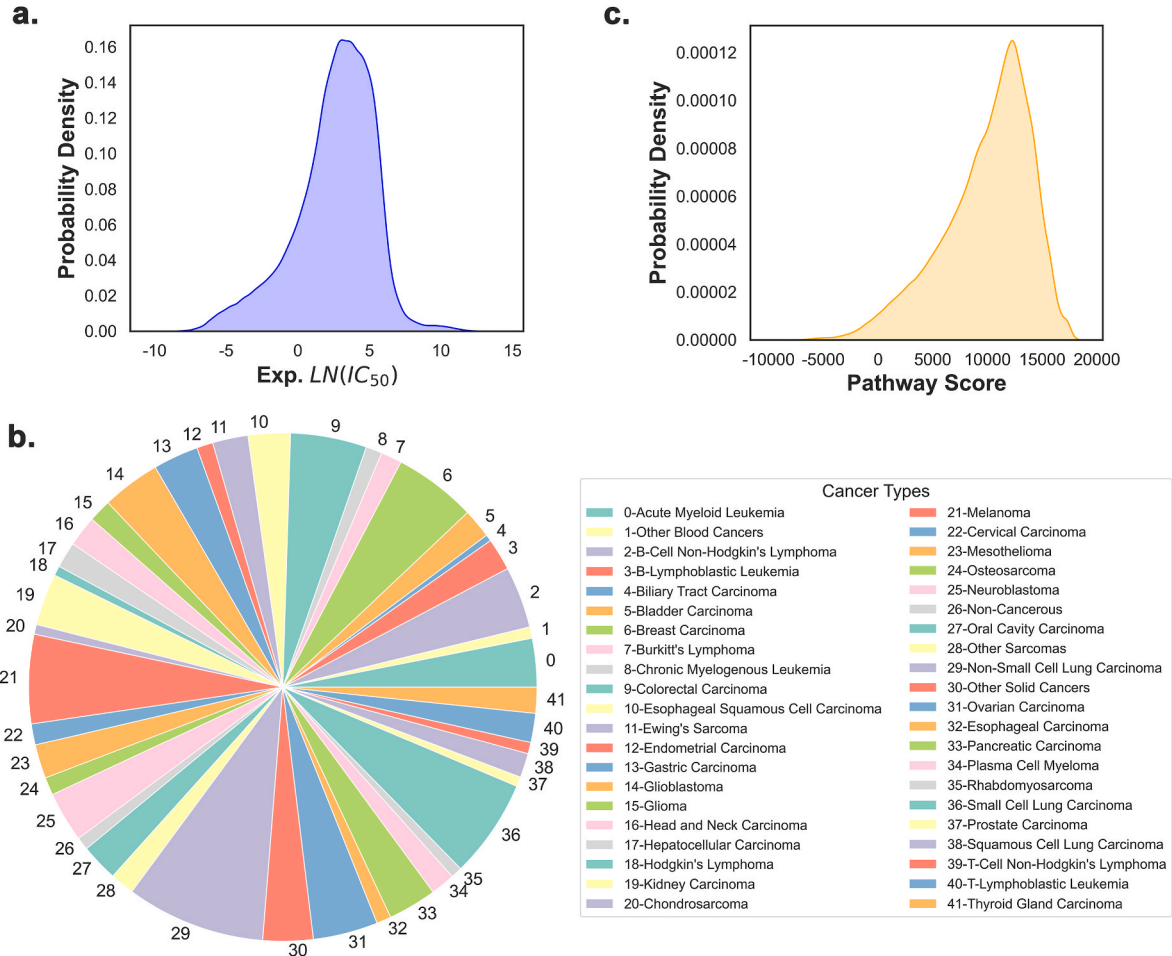


Fig. 2. Analysis of drug sensitivity profiles used in this study. (a) The distribution of drug response $LN(IC_{50})$ values. (b) Map of cancer cell family categories in our GDSC datasets, including 42 cancer types. (c) The distribution of ssGSEA pathway enrichment scores.

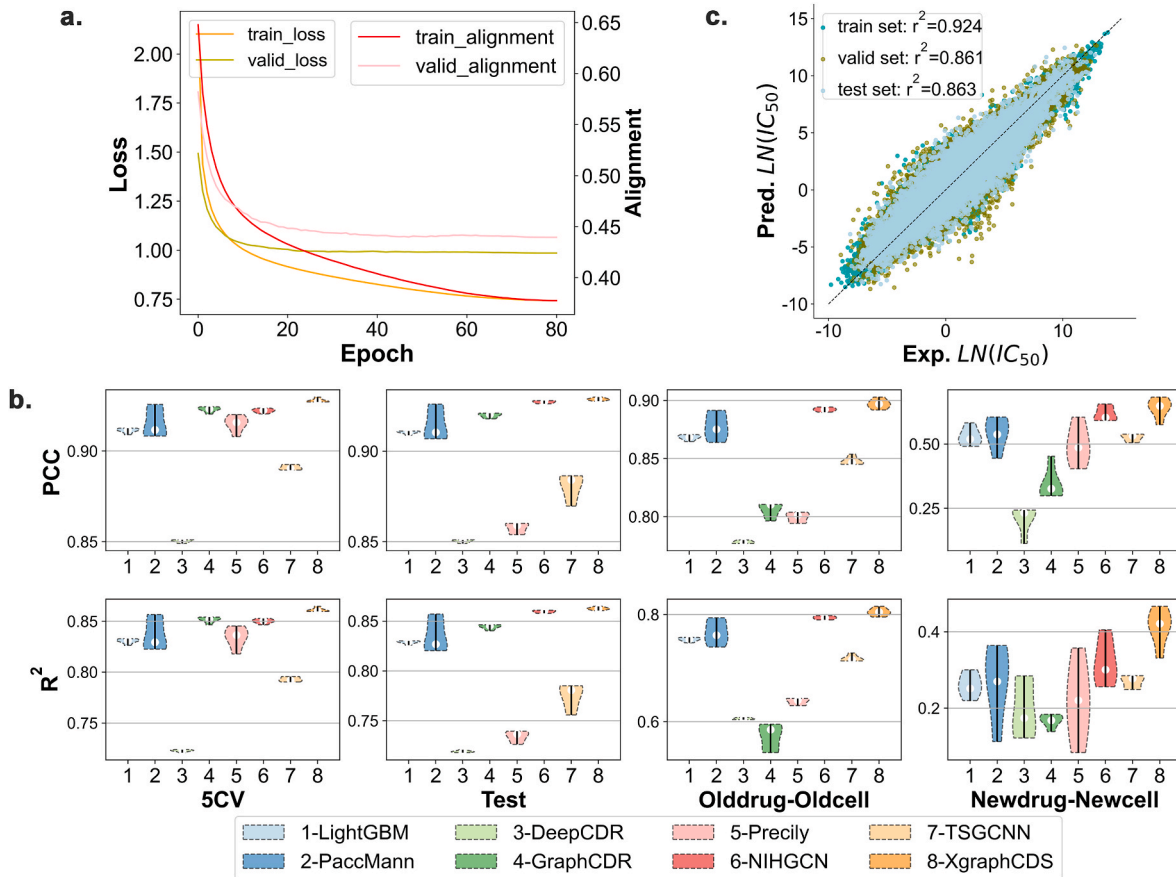


Fig. 3. The performance of XGraphCDS in different experimental configurations on regression task. **(a)** Curves of loss versus alignment during pre-training of the model. **(b)** Violin plot of XGraphCDS and baseline performance in the regression task. **(c)** A linear scatter plot of the training, validation, and test sets in a regression model. The closer the scatter is to a straight line, the better the approximation of the model predictions to the true values is.

2.2.6. Interpretability analysis

Lack of interpretability is another issue of concern for machine learning and deep learning, often referred to as “black box models” [45]. Using gradients to explain neural network models is a direct approach, but it could suffer from gradient saturation. To address this problem, Sundararajan and his colleagues proposed the concept of integrated gradients (IG), which defines the contribution of an input feature as the integral of the path from the baseline to that feature [46]. In this work, we set the baseline as $x' = 0$ and compute the integrated gradients of each input feature x (atom or gene pathway) through backpropagation of the XGraphCDS network to obtain explanations for the model. As shown in Fig. 1d, node weights can be visualized as atomic heat maps superimposed on the molecular graph, and gene pathway weights are represented as importance matrices. The specific calculation formula is as follows:

$$IG_x = (x - x') \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x} d\alpha \quad (2)$$

where x is the input feature (atom or gene pathway), x' is the baseline, α is the interpolation parameter, $x' + \alpha(x - x')$ denotes the interpolation process from baseline x' to original input x , ∂F is the model function.

2.3. Model evaluation experiments

2.3.1. Assessment metrics

In this study, we constructed both quantitative (regression) and qualitative (binary classification) models for drug sensitivity. The regression model, its performance was mainly evaluated by four

evaluation metrics: PCC (Pearson correlation coefficient), R^2 (coefficient of determination), MAE (mean absolute error), and RMSE (root mean square error). For the classification model, due to the imbalanced dataset, we mainly used five evaluation metrics: balanced accuracy (BACC), the area under the curve (AUC), the area under the precision-recall curve (PRAUC), accuracy (ACC), and F1 score. In addition, we used two metrics, Uniform and Alignment [47], to evaluate the learning ability of the self-supervised pre-training method based on contrastive learning. The Alignment is only related to positive pairs, aiming to bring the feature vectors of positive pairs closer together, making them more similar in the embedding space. The Uniform is only related to negative pairs, aiming to preserve as much information as possible to better distinguish between positive and negative samples.

2.3.2. Baselines

To evaluate the performance of the XGraphCDS model, we compared our method with the state-of-the-art baseline models, including traditional machine learning models and advanced deep learning models. We used 5-fold cross-validation to evaluate them on the same validation set, test set, and external validation set.

- **LightGBM** [48] is a distributed gradient-boosting framework. We use the gene pathways of cell lines and drug molecular fingerprints as the input for the machine-learning models. Similarly, considering that too many input features may cause overfitting, we use principal component analysis (PCA) [49] to reduce the cell line and drug representations to a feature vector of length 256 (same as XGraphCDS). All parameters are determined by the grid random

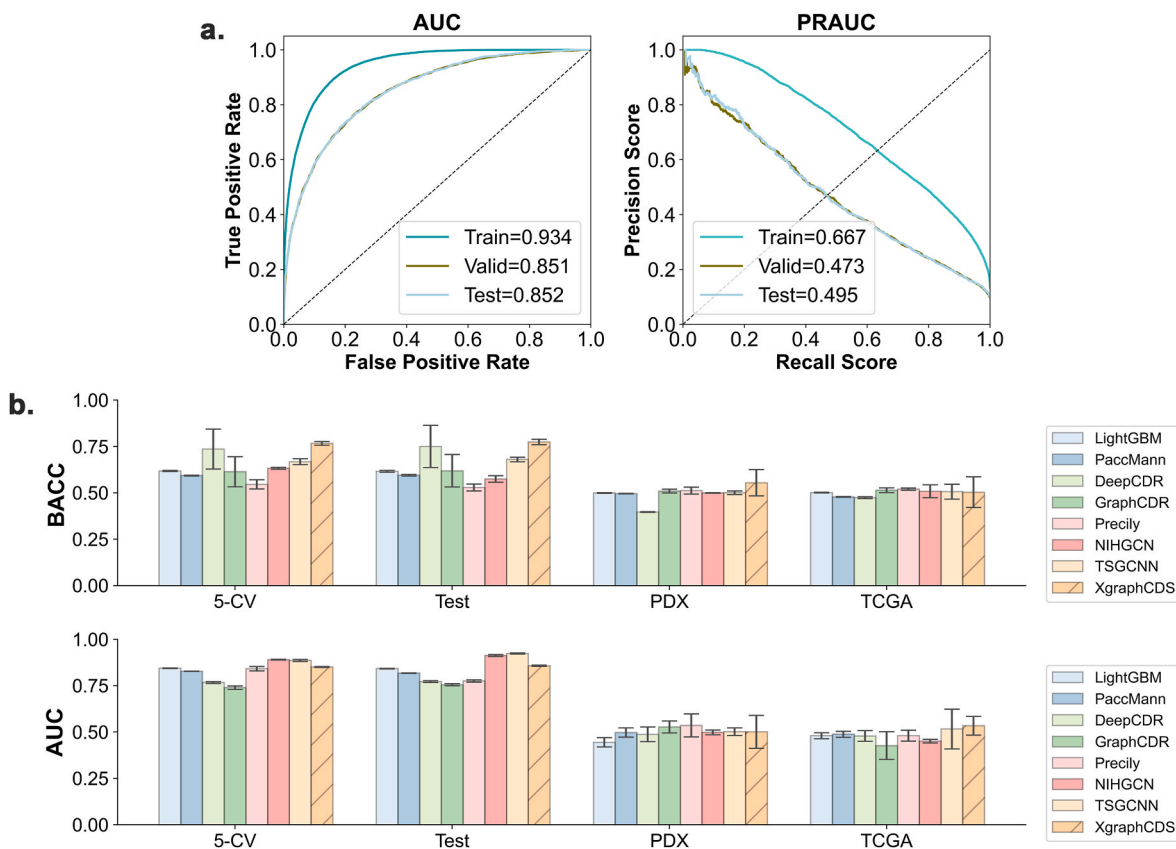


Fig. 4. The performance of XGraphCDS in different experimental configurations on binary task. **(a)** The receiver operating characteristic (ROC) curve and precision–recall (PR) curve were reported to give its binary label (resistance/sensitivity). **(b)** Histogram of XGraphCDS and baseline performance in the classification task.

search algorithm and implemented using the Scikit-learn library [50].

- **DeepCDR** [16] is a hybrid graph convolutional network that integrates multi-omics profiles of cancer cells and explores the intrinsic chemical structure of drugs for prediction.
- **PaccMann** [17] uses a multimodal attention-based neural network to predict the sensitivity of anti-cancer compounds. This method currently provides an open web service at <https://ibm.biz/paccmann-aas>.
- **GraphCDR** [11] is a GNN for drug response prediction built upon multiple omics features of cancer cell lines, and drug chemical structures. It also introduced a contrastive learning task as a regularizer to enhance generalization capability.
- **Precily** [18] is a predictive modeling method that uses gene expression data to infer cancer drug response, demonstrating the benefits of concatenating pathway scores with drug descriptors as features.
- **NIHGCN** [12] proposed a neighborhood interaction-based heterogeneous graph convolutional network approach for end-to-end anti-cancer drug response prediction. The input consists of gene expression of cell lines and molecular fingerprints of drugs.
- **TSGCNN** [13] is a deep learning algorithm based on two-space graph convolutional neural networks for predicting the drug response.

3. Results and discussion

3.1. Statistics and analysis of drug sensitivity profiles

For the *in vitro* dataset, after data processing, we obtained 339,017 and 187,849 cell line-drug interaction pairs to establish quantitative (maximum half-inhibitory concentration $LN(IC_{50})$ values, Fig. 2a) and

qualitative (sensitivity or resistance) of drug sensitivity respectively, covering 42 cancer types (Fig. 2b). For the *in vivo* dataset, we collected 3,039 pairs of cancer patient-drug interaction data, including 141 drugs and 29 types of cancer. After performing single sample Gene Set Enrichment Analysis (ssGSEA) [30] on high-dimensional gene expression profile data, we obtained a total of 2998 gene pathway scores as input features representing cancer samples. The pathway enrichment score distribution is presented in Fig. 2c. The high-dimensional nature of genomic feature matrices presents significant challenges, including increased computational complexity, over-fitting, and difficulties in analyzing and interpreting the data. Some work only selects some representative genes for modeling, but this may lead to the loss of useful information. Our approach not only retained their biological significance but also expanded the input mode to improve application value. Because gene expression in each cell or tissue may be different, the dimensionality of the input data may change due to missing data, which can make fixed-dimension models inaccessible.

3.2. *In vitro* prediction model evaluation of drug sensitivity

3.2.1. Pre-training model assessment

Observing Table S1, our drug sensitivity dataset only includes 400 drugs, and the narrow chemical space distribution may severely limit the model's application domain. Therefore we designed a self-supervised pre-training strategy based on contrastive learning to supplement drug information. During the training process, we checked the GNN_{drug} pre-training module of XGraphCDS every epoch and visualized the changes in the loss, Alignment, and Uniformity indicators in Fig. 3a and Fig. S4. It can be observed that our pre-training model continuously converged with increasing iterations, and the gains in Alignment and Uniformity were accumulated, ultimately optimizing Alignment to

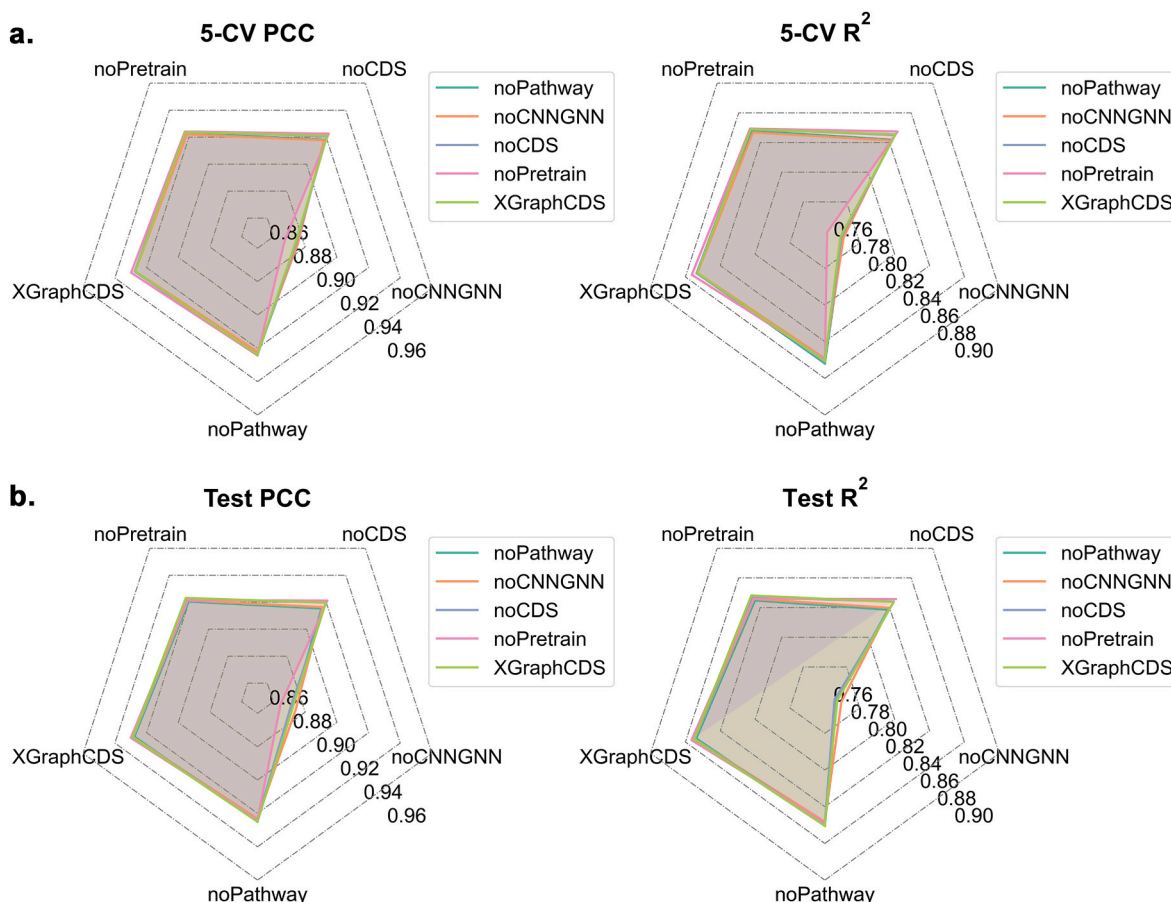


Fig. 5. Ablation experiment results. (a) Radar plot of PCC and R^2 values for five-fold cross-validation (5-CV). (b) Radar plot of PCC and R^2 values for the test set.

0.439 and Uniformity to -3.94 . This indicated that our pre-training strategy not only learned features that maximize the similarity between positive samples but also preserved the chemical information contained in the original molecular graph to the fullest extent.

3.2.2. Continuous drug sensitivity prediction

In this section, we designed a series of experiments to observe whether XGraphCDS can help infer continuous drug sensitivity. We summarized the prediction accuracy of all models in Fig. 3b and Table S4. Overall, the deep neural network models performed significantly better than traditional machine learning algorithms, especially the deep models based on GNN networks, which may be because graph-based algorithms are better at capturing the intrinsic chemical structural information of drugs. XGraphCDS obtained PCC scores of 0.927 and 0.929, R^2 scores of 0.861 and 0.863, RMSE scores of 1.045 and 1.030, and MAE scores of 0.776 and 0.772 on 5-CV and test sets, respectively (Fig. 3c). Compared with seven competing methods, XGraphCDS demonstrated excellent performance in drug response prediction in regression experiments, with prediction accuracy exceeding most baselines on each dataset and reaching a similar level as the best-performing baseline NIHGCN. Additionally, we conducted two independent tests to further evaluate the model's performance on unknown interactions. In predicting missing cell interactions within the network, XGraphCDS achieved the highest PCC score (0.898) and R^2 score (0.807), as well as the lowest RMSE (1.152) and MAE score (0.846), compared to other baselines. In studying the ability of drug induction outside the network, the performance of all methods decreased significantly, which may be due to the imbalance in the number of cell lines and drug nodes and the limited known chemical space of drugs. However, XGraphCDS still achieved the best results, with a PCC score of

0.649, an R^2 score of 0.421, and the lowest MAE and RMSE scores of 1.309 and 1.726. This may be attributed to our self-supervised pre-training strategy based on contrastive learning to enhance the chemical information of the drug encoding sub-network and the cell line representation method based on pathway enrichment scores. The above analysis indicated that XGraphCDS could simultaneously address two complementary goals of drug sensitivity prediction models, generalizing to unseen cancer cell lines (precision medicine system) and unseen compounds (drug discovery system).

3.2.3. Binary drug sensitivity prediction

We constructed a binary classification model for drug sensitivity based on the reported maximum screening concentration of the drug [20,21]. After filtering out drugs without thresholds, we were left with only 18,256 positive samples (indicating drug sensitivity to the cell line). The dataset (Table S5) was highly imbalanced. Despite this, XGraphCDS still outperformed most of the baselines (Fig. 4, Table S6, and Table S7). Our model achieved an AUC value of 0.851 and a BACC value of 0.767, indicating a high accuracy in predicting drug sensitivity and good performance on the imbalanced dataset. After training the model on the aforementioned *in vitro* cell line data, we conducted blind tests on two independent datasets, PDX and TCGA, to evaluate the generalizability of XGraphCDS to unseen data. It is important to note that there is also an imbalance of positive and negative data in the PDX data (20 positive samples and 162 negative samples), and the chemical space of drugs in the TCGA dataset is different from that in the GDSC training set (Fig. S5). Nevertheless, XGraphCDS still performed well on both external validation sets and was competitive among numerous baseline models, with BACC scores of 0.501 and 0.534, AUC scores of 0.501 and 0.534, and PRAUC scores of 0.172 and 0.616, respectively.

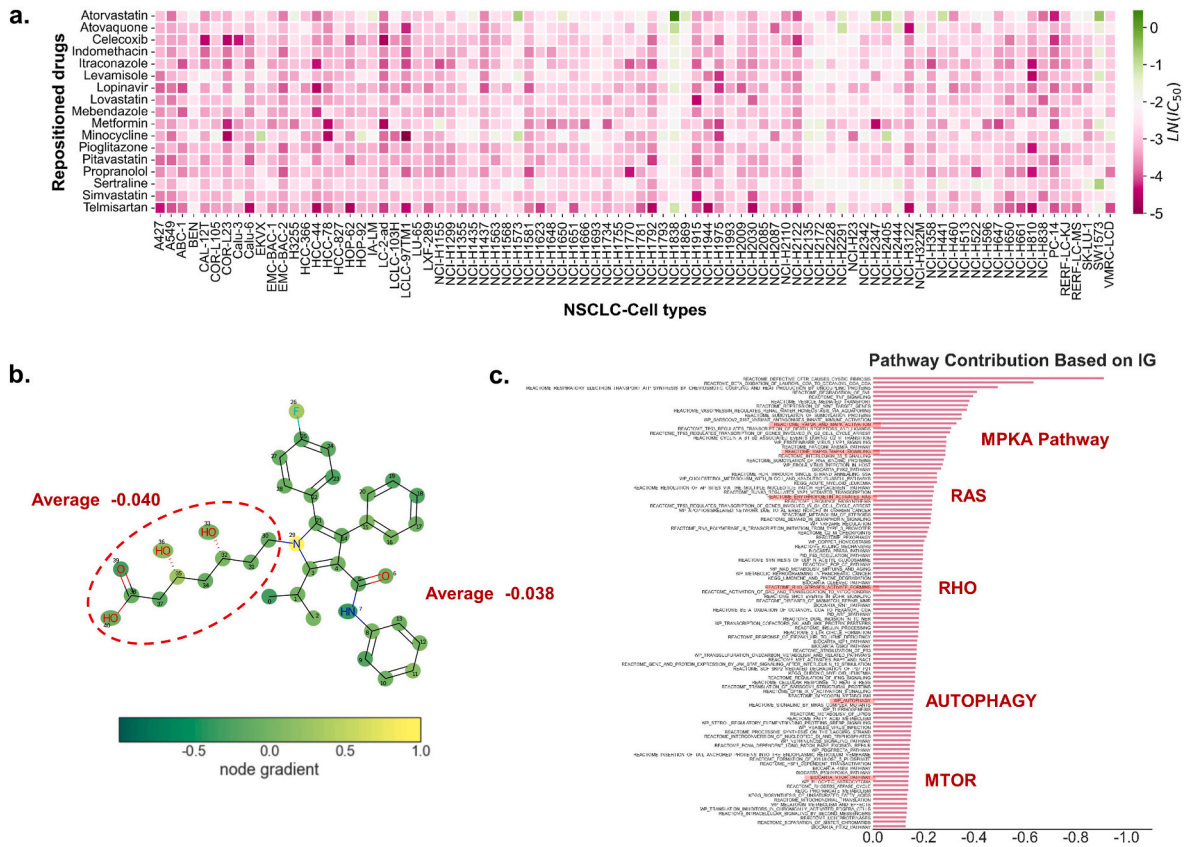


Fig. 6. The results and interpretation of drug repositioning by XGraphCDS model for NSCLC. **(a)** Heatmap of the predicted $\text{LN}(\text{IC}_{50})$ of cell sensitivity for 17 NSCLC repositioning drugs, with redder colors being more sensitive. **(b)** Visualization of the atom weights of Atorvastatin. The darker the color, the higher the value of the atomic weight. **(c)** Contribution ranking of top 100 gene pathways.

We found that XGraphCDS outperforms other existing methods, demonstrating the highest BACC metric. This could be attributed to our model's training methodology utilizing contrastive learning and Focal Loss, effectively capturing the relationship between cancer samples and drug sensitivity, thus enhancing stability and predictive capability. Additionally, while NIHGCN and TSGCNN exhibited good performance in terms of AUC, there was a significant decrease in PRAUC, indicating lower precision at high recall rates. In contrast, our XGraphCDS model demonstrated consistent performance and maintained good metrics throughout the evaluation process. This suggests that the XGraphCDS model strikes a balance between accuracy and recall rates at different thresholds, thereby providing a more comprehensive predictive ability for cancer drug sensitivity. Overall, the stable performance of the XGraphCDS model across multiple performance metrics such as BACC and AUC further validates its superiority and reliability in capturing information on drug and cancer cell interactions.

3.2.4. Ablation experiments

To investigate the magnitude of various components in XGraphCDS, we trained the XGraphCDS regression model de novo for model ablation analysis and designed the following variants: i. no pathway enrichment (noPathway), with cell line input as hallmark genes; ii. deletion of the CNN_{gene} and GNN_{drug} channels (noCNNGNN), using only the cell line pathway score and ECFP4 molecular fingerprint (1024 bits) inputs; iii. removal of the CDS hetero-aggregate graph (noCDS), with direct concatenation for prediction; iv. deletion of the self-supervised pre-training module (noPretrain), with no drug information enhancement.

The 5-CV and test results of all variant models trained on the GDSC dataset are summarized in Fig. 5. XGraphCDS achieved the largest enclosed area compared to its variants, indicating superior performance. We observed that the straightforward adoption of signature genes to

represent cell lines led to a decrease in model prediction performance, which was consistent with our hypothesis that this practice would lose partial vital information. We also noted that the performance of the model variants without the CNN_{gene} and GNN_{drug} channels, and without the CDS hetero-aggregation structure, was lower than that of the original XGraphCDS model, indicating that these three modules contributed to the prediction. The CNN_{gene} and GNN_{drug} channels effectively extracted potential information about the cell lines and drugs. By constructing a hetero-graph network that fused the hidden vectors of both after feature extraction, the imbalance of the cell line and drug quantities was weakened compared to direct concatenation, which is beneficial for capturing the inherent mechanism of drug response. In addition, because of our pre-training strategy, the PCC and R^2 scores of XGraphCDS were increased by 1.2% and 1.1%, respectively. This suggests the rationality of our comparative learning task design, where GNN_{drug} continuously accumulated chemical knowledge of PubChem molecules during pre-training to alleviate the narrow drug chemistry space in the original data. Through ablation analysis, we could conclude that the XGraphCDS framework combines multiple strategies and achieves higher performance than its variants, indicating that it successfully improves CDS prediction using a hetero-graph convolutional neural network based on cell line pathway scores and drug structures.

3.3. Model applicability and interpretability

3.3.1. Drug repurposing for non-small cell lung cancer

Lung cancer is the second leading cancer, with the majority (80%) of lung cancer cases falling into the non-small cell lung carcinoma (NSCLC) subtype [51]. As proof of the anti-cancer drug repurposing potential of XgraphCDS, 86 NSCLC cell lines were used for drug screening. We collected an additional 17 drugs that have potential anti-NSCLC activity

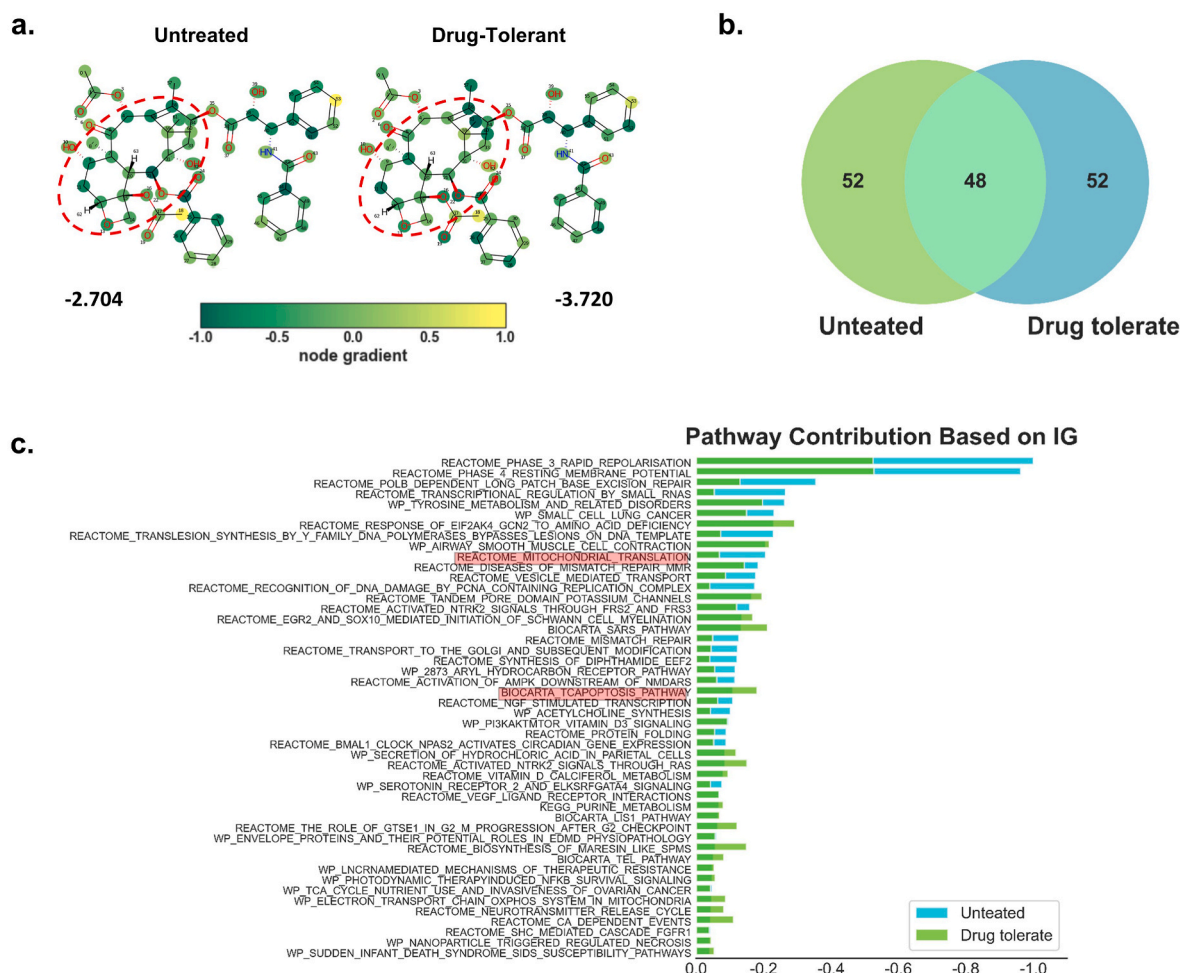


Fig. 7. The predicted drug sensitivity of paclitaxel was assessed in two MDA-MB-231 cells. **(a)** Visualization of the atom weights of paclitaxel in both cell types. The key pharmacophores are shown in red circles. **(b)** Venn diagram of the top 100 gene pathways differential profiles of untreated and drug-tolerant cells. **(c)** The comparison chart of the 48 shared pathways in the Venn diagram.

in addition to their original drug indications. It should be noted that these data pairs are not included in our training set, as shown in Table S8. We predicted the interactions of these 1462 drug-cell line pairs using both quantitative and qualitative models. For the quantitative model, $\text{LN}(\text{IC}_{50}) < -2.0$ was used to define a sensitive anti-cancer drug response, based on previous studies [52,53]. As shown in Fig. 6a and Fig. S6, most drugs were predicted to be sensitive in over 80% of NSCLC cell lines, which validates the drug repositioning capabilities of our model. Additionally, the model summarized the potential gene pathways and pharmacophores of these drugs for NSCLC treatment. Using the anti-hyperlipidemic drug Atorvastatin as an example, the key pharmacophore was identified as dihydroxy pentanoic acid [54]. According to the atom visualization results of the model, the sum of the atom weights in this region was negative (reducing IC_{50} and enhancing inhibitory activity), which was highlighted in green (Fig. 6b and Table S9). We also validated the importance of these two regions using local mean comparison. If the absolute value of the average atomic weight in the region is greater than that outside the region, it is considered a key pharmacophore [55]. It has been reported that Atorvastatin can exert anti-tumor effects by inhibiting the Akt/mTOR pathway and activating the MAPK pathway while inhibiting cell growth stimulant proteins Ras, Rac, and Rho to induce apoptosis of tumor cells [51,56]. Atorvastatin can also induce apoptosis of NSCLC by stimulating autophagy to produce reactive oxygen species [57,58]. As shown in Fig. 6c, these pathways were among the top 100 pathways in the model's list of sensitivity to gene pathways. It should be noted that this is just one

case study, and our model can be used for drug screening and explanatory analysis of any type of cancer.

3.3.2. Paclitaxel resistance analysis

To investigate drug tolerance-related discontinuation, we found an instance for validation. In Lee et al.'s study, metastatic breast cancer cells (MDA-MB-231) exposed to paclitaxel for five days resulted in the majority of the cells dying, with a small population of remaining cells developing into drug-resistant clones [59]. These resistant cells exhibited a gene expression profile similar to that of untreated cells and were more sensitive to paclitaxel. We predicted this scenario and found that the model yielded a similar trend to the experiment, where the drug response values of the resistant cell population were lower ($\text{LN}(\text{IC}_{50}) = -3.720$) compared to the untreated MDA-MB-231 cells ($\text{LN}(\text{IC}_{50}) = -2.704$). We then analyzed this situation by the XGraphCDS model. First, we visualized the atom weights of paclitaxel (Fig. 7a) and found that most of the atoms had a negative effect, with increased drug sensitivity, especially in the resistant cell line. The taxane scaffold of paclitaxel was found to be necessary for anticancer activity [60,61], and we confirmed this by summing the atom weights in this region (negative values, sensitive) and comparing them with the local averages (Table S10). Additionally, our model provided a ranking of the top 100 gene pathways sensitive to both scenarios, and we found that 50% of the pathways overlapped according to a Venn diagram (Fig. 7b). Pathways related to mitosis and apoptosis were present in both lists (Fig. 7c) due to paclitaxel's action of targeting the microtubules to interfere with spindle

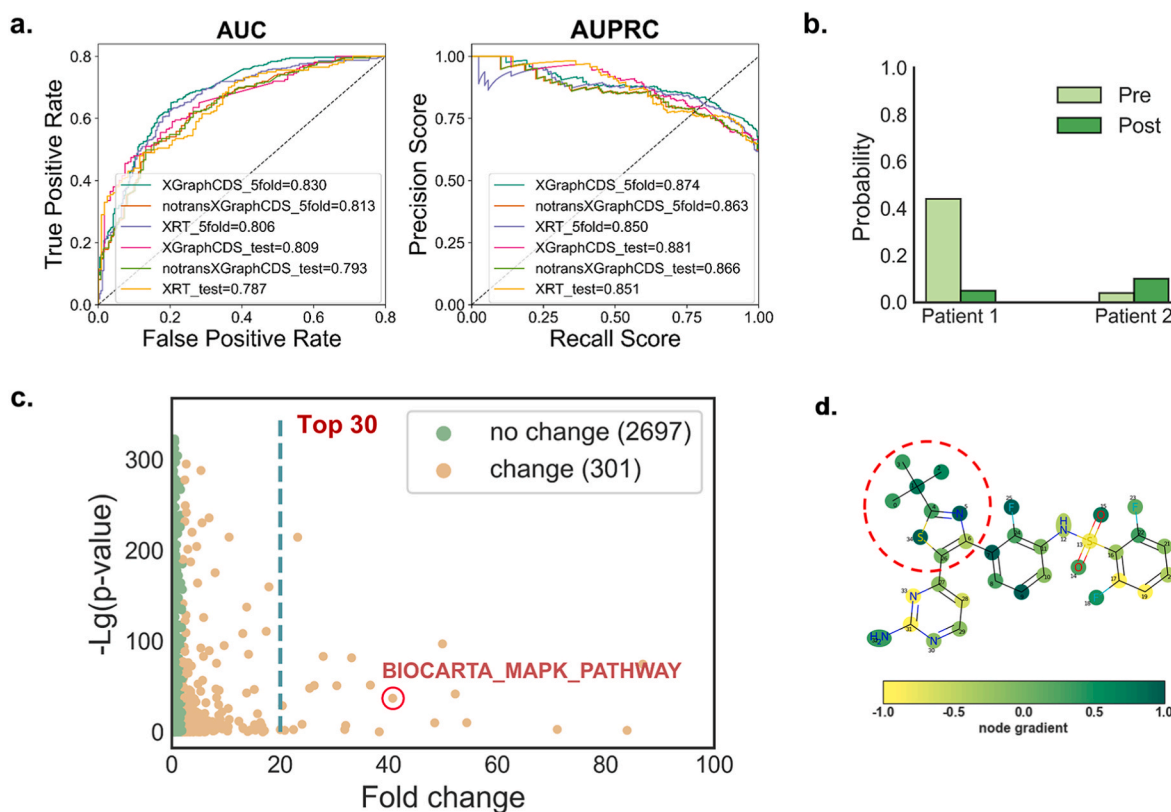


Fig. 8. A clinical interpretation assessment *in vitro* prediction model. (a) The receiver operating characteristic (ROC) curve and precision–recall (PR) curve were reported to give its binary label (resistance/sensitivity). (b) Bar graph showing the predicted sensitivity probabilities for the drug dabrafenib pre- and post-dosing in patient 1 and patient 2. (c) Scatter plot of pathway difference analysis based on XGraphCDS model, the 30 pathways with the highest variance are shown on the right side of the blue dashed line. (d) Visualization of the atom weights of Dabrafenib. The darker the color, the higher the value of the atom weight.

formation during mitosis and cause cell cycle arrest [62]. Notably, microtubule motor activity, cell surface signaling, and the IGF1R pathway showed significant effects in drug-tolerated cells (Fig. S7), while their effects on pre-treatment cells were small. This could explain why the resistant cells are more sensitive to paclitaxel. The results obtained from our model were consistent with Lee et al.'s study, indicating that the XGraphCDS framework could not only accurately predict drug discontinuation but also help elucidate the reasons for drug resistance, demonstrating its reliability and practicality.

3.4. *In vivo* prediction model evaluation and clinical application

Although *in vitro* prediction models we previously constructed have some predictive power for the activity of compounds *in vivo*, they still require comprehensive evaluation in conjunction with *in vivo* experimental data [63]. Therefore, based on the XGraphCDS framework, we constructed separate *in vivo* models for complete/partial response and non-response to drugs in TCGA patients. Given the diversity of cancer genomics and compound structures, the collected *in vivo* data are insufficient. Therefore, we continued to train the TCGA dataset based on the parameters of the trained GDSC *in vitro* model, transferred all the parameters in the GDSC model (source domain), and validated and optimized it on TCGA *in vivo* data (target domain). We hope that the XGraphCDS model can learn and predict almost any sample-drug pair combination. As shown in Fig. 8a, the XGraphCDS model exhibited excellent predictive ability, with AUC values of 0.830 and 0.808 and corresponding PRAUC values of 0.874 and 0.877 calculated on the 5-CV and test sets, respectively. After transfer learning, the AUC and PRAUC scores on the 5-CV validation set increased by 2% and 1%, and those on the test set increased by 2.5% and 1.8%, demonstrating the necessity of our transfer learning strategy. In addition, we also presented

experimental results of Extremely Randomized Trees (XRT), which Chawla et al. [18], validated as the best-performing method among 34 models in TCGA *in vivo* data. It could be observed that our XGraphCDS model scores were better than XRT. Combining *in vitro* and *in vivo* data through transfer learning can improve the efficiency and accuracy of drug development, help develop personalized treatment plans for cancer patients, and reduce the need for animal experiments and clinical trials.

Then, we applied the XGraphCDS *in vivo* prediction model to an independent dataset to evaluate its clinical applicability. The dataset was derived from the study by Wagle et al. which included RNA-seq gene expression profiles and clinical response information of melanoma patients in pre- and post-treatment with RAF inhibitor (Dabrafenib) [64]. As shown in Fig. 8b, our model predicted that Patient 1 had a higher probability of drug sensitivity to Dabrafenib in pre-treatment while Patient 2 was always resistant, consistent with their treatment histories. Differential analysis (DA) based solely on gene expression levels might lose some latent spatial information, and we used the XGraphCDS model to perform DA on the gene pathways of Patient 1 in pre- and post-treatment, identifying 301 pathways with significant changes (fold change > 2 and p-value < 0.05). It was noteworthy that Wagle et al. reported that RAF resistance mechanisms might be due to changes in the mitogen-activated protein kinase (MAPK) pathway caused by MEK2Q60P mutations, BRAF splicing isoforms, and BRAF amplification after drug administration. As observed in Fig. 8c and Fig. S8, the BIO-CARTA_MAPK_PATHWAY weight increased 40-fold, ranking 16th. In addition, our model also provided the atomic weights of Dabrafenib (Fig. 8d and Table S11). According to reports by Rheault et al. the buthylthiazole core is a key pharmacophore required for the effective inhibition of B-Raf^{V600E} cells by Dabrafenib [65]. In our model analysis, most of the atoms in the buthylthiazole core were highlighted in green and proved to be significant by local mean comparison. This successful

example demonstrates that our XGraphCDS *in vitro* model can help in developing personalized dosing regimens and overcome specific resistance mechanisms in clinical settings.

4. Conclusion

Due to the inherent heterogeneity and genetic variations in cancer, predicting cancer drug sensitivity is an important issue in precision medicine research. On the one hand, clinicians need to choose the most beneficial and least toxic treatment plans. On the other hand, drug researchers also need to design highly specific anti-cancer drugs. In this study, we developed a novel framework called XGraphCDS by integrating cancer genomics data and clinical data. It consists of a unified heterogeneous network and multiple sub-networks, with molecular graphs and gene enrichment scores serving as inputs representing drugs and cancer samples, respectively. XGraphCDS utilizes this input information for training and optimization to predict cancer drug sensitivity, providing deeper insights and understanding. The robustness and accuracy of the model have been validated through 5-fold cross-validation and different external validation sets, enabling predictions for unknown drug-cancer sample pairs. The contrastive learning-based pre-training module contributes significantly. In addition, the XGraphCDS method, with its interpretability, not only accurately predicts drug sensitivity but also provides resistance mechanism analysis, aiding in the identification of biomarkers. Finally, we employed transfer learning strategies to successfully extend *in vitro* experimental data to *in vivo* datasets and constructed a separate *in vivo* prediction model which also achieved encouraging predictive results and has been validated for its clinical relevance.

In conclusion, our research has both theoretical and practical implications that contribute to the field of precision medicine. For clinicians, our model aids in the selection of personalized treatment plans that maximize efficacy while minimizing toxicity. It assists in choosing the most beneficial drugs for individual patients, improving patient outcomes in precision medicine. Drug researchers can benefit from our findings by using the XGraphCDS framework to design highly specific anti-cancer drugs. Additionally, our model enables drug repositioning, identifying potential new applications for existing drugs, and reducing the time and cost required for traditional drug development processes. Furthermore, the identification of biomarkers for drug sensitivity through our model assists in tailoring treatments and developing targeted therapies. The XGraphCDS framework enhances our understanding of drug sensitivity prediction and provides valuable insights for personalized treatment decisions, drug discovery, drug repositioning, and biomarker identification.

It is worth noting that although XGraphCDS has achieved promising results in predicting cancer drug sensitivity, it also has certain limitations. Firstly, the practicality of the XGraphCDS model is still to be further validated due to the limited cell types and drugs in the current dataset. Our model also has the potential to be trained on a larger dataset, which could lead to higher performance. Integrating information from multiple publicly available cancer drug screening databases can help build a more powerful and versatile model. However, experimental inconsistencies and noise across different datasets hinder efforts to achieve this goal. Secondly, XGraphCDS currently only considers the molecular and genetic features of cells, and including other relevant background information such as clinical factors may improve its performance. Future work can focus on addressing these limitations to further enhance the utility of XGraphCDS in practical applications.

Author contributions

Guixia Liu and Yun Tang supervised the research project. Yimeng Wang and Long Chen collected data. Yimeng Wang designed and implemented the XGraphCDS method. Yimeng Wang constructed the models and evaluated the performance. Yimeng Wang, Yaxin Gu, and

Xinxin Yu analyzed the prediction results. Yimeng Wang, Weihua Li, Guixia Liu, and Yun Tang wrote the manuscript. All authors read the manuscript and approved the final version.

Data availability

All the attribution codes, trained models, and datasets are available on our Git Hub (<https://github.com/Yimeng-Wang/XGraphCDS>).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The current research was supported by financial support from the National Key Research and Development Program of China (Grant 2019YFA0904800), the National Natural Science Foundation of China (Grants 82173746, 82273858, and 82273930), and Shanghai Frontiers Science Center of Optogenetic Techniques for Cell Metabolism (Shanghai Municipal Education Commission).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2023.107746>.

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